

EXPERIMENT NO. 9

Title

Connect Spring Boot to MySQL/PostgreSQL using Spring Data JPA and Perform CRUD Operations

Aim

To connect a Spring Boot application with MySQL/PostgreSQL database using Spring Data JPA and perform CRUD (Create, Read, Update, Delete) operations.

Software Requirements

Operating System: Windows / Linux / macOS

Programming Language: Java (JDK 17 or above)

Framework: Spring Boot 3

Database: MySQL / PostgreSQL

Tools: IntelliJ IDEA / Eclipse / VS Code, Postman

Hardware Requirements

Processor: Intel i3 or higher

RAM: Minimum 4 GB

Hard Disk: 20 GB (minimum)

Introduction

Spring Data JPA simplifies database access in Spring Boot applications by reducing boilerplate code. It provides repository support to perform CRUD operations using JPA and ORM concepts.

Theory

JPA (Java Persistence API) is a specification for object-relational mapping. Spring Data JPA provides built-in repository interfaces such as `JpaRepository` to easily perform CRUD operations on relational databases.

Algorithm

1. Create a Spring Boot project
2. Configure database connection in `application.properties`
3. Create Entity class with JPA annotations
4. Create Repository interface extending `JpaRepository`
5. Create REST Controller for CRUD operations
6. Run the application and test APIs using Postman

Program Code

```
// application.properties
```

```
spring.datasource.url=jdbc:mysql://localhost:3306/studentdb
spring.datasource.username=root
spring.datasource.password=root
spring.jpa.hibernate.ddl-auto=update
```

```
// Entity Class
@Entity
class Student {
    @Id @GeneratedValue
    private Long id;
    private String name;
}

// Repository Interface
interface StudentRepository extends JpaRepository<Student, Long> {}

// Controller
@RestController
@RequestMapping("/students")
class StudentController {
    @Autowired
    private StudentRepository repo;

    @PostMapping
    public Student save(@RequestBody Student s) { return repo.save(s); }

    @GetMapping
    public List<Student> getAll() { return repo.findAll(); }
}
```

Output

CRUD operations successfully performed using REST APIs.
Data stored and retrieved from MySQL/PostgreSQL database.
(Attach screenshots of Postman responses)

Learning Outcomes

Students will be able to integrate Spring Boot with relational databases using Spring Data JPA and perform CRUD operations.

Conclusion

Thus, Spring Boot was successfully connected to MySQL/PostgreSQL database using Spring Data JPA and CRUD operations were performed.